

50.007 Machine Learning, Spring 2025
Homework 2

Question 2: Soft-Margin SVM with Kernel Methods

(a) Primal and Dual Problems for Soft-Margin SVM with Kernel Methods

Given the training set

$$\{(x_i, y_i)\}_{i=1}^n, \quad y_i \in \{-1, +1\},$$

and the feature mapping $\phi(x)$ associated with the kernel

$$K(x, x') = \phi(x) \cdot \phi(x').$$

In the soft-margin formulation we introduce slack variables $\xi_i \geq 0$ (one per training example) to allow misclassifications and a regularization parameter $C > 0$ to control the penalty.

Primal Problem in the ϕ -space:

$$\begin{aligned} \min_{\theta, \theta_0, \xi} \quad & \frac{1}{2} \|\theta\|^2 + C \sum_{i=1}^n \xi_i \\ \text{subject to} \quad & y_i (\theta \cdot \phi(x_i) + \theta_0) \geq 1 - \xi_i, \quad i = 1, \dots, n, \\ & \xi_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

Lagrangian Function: Introduce Lagrange multipliers

$$\alpha_i \geq 0, \quad i = 1, \dots, n,$$

for the constraints

$$y_i (\theta \cdot \phi(x_i) + \theta_0) \geq 1 - \xi_i,$$

and Lagrange multipliers for the non-negativity constraints $\xi_i \geq 0$:

$$\mu_i \geq 0, \quad i = 1, \dots, n.$$

Then the Lagrangian function is

$$L(\theta, \theta_0, \xi, \alpha, \mu) = \frac{1}{2} \|\theta\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i [1 - \xi_i - y_i (\theta \cdot \phi(x_i) + \theta_0)] - \sum_{i=1}^n \mu_i \xi_i.$$

To satisfy KKT Conditions (Stationarity), we can derive:

1. With respect to θ :

$$\frac{\partial L}{\partial \theta} = \theta - \sum_{i=1}^n \alpha_i y_i \phi(x_i) = 0 \implies \theta = \sum_{i=1}^n \alpha_i y_i \phi(x_i).$$

2. With respect to θ_0 :

$$\frac{\partial L}{\partial \theta_0} = - \sum_{i=1}^n \alpha_i y_i = 0 \implies \sum_{i=1}^n \alpha_i y_i = 0.$$

3. With respect to ξ_i :

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \mu_i = 0 \implies \mu_i = C - \alpha_i.$$

Since $\mu_i \geq 0$, it follows that

$$0 \leq \alpha_i \leq C \quad \text{for } i = 1, \dots, n.$$

Substituting the optimality conditions back into the Lagrangian function and eliminating θ , θ_0 , and ξ , the objective becomes a function of α only. Then we can get the dual problem:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j K(x_i, x_j) \\ \text{subject to} \quad & \sum_{i=1}^n \alpha_i y_i = 0, \\ & 0 \leq \alpha_i \leq C, \quad i = 1, \dots, n. \end{aligned}$$

(b) Optimal Parameters and Support Vectors

Optimal $\hat{\theta}$: From the stationarity condition with respect to θ , the optimal parameter vector in the feature space is given by

$$\hat{\theta} = \sum_{i=1}^n \hat{\alpha}_i y_i \phi(x_i).$$

Optimal $\hat{\theta}_0$: For any training example that is a *support vector* on the margin (with $0 < \hat{\alpha}_i < C$) and typically satisfying $\xi_i = 0$, the complementary slackness condition implies

$$\hat{\alpha}_i \left[1 - y_i \left(\hat{\theta} \cdot \phi(x_i) + \hat{\theta}_0 \right) - \xi_i \right] = 0.$$

Since $0 < \hat{\alpha}_i < C$, the classification constraint $1 - y_i \left(\hat{\theta} \cdot \phi(x_i) + \hat{\theta}_0 \right) - \xi_i = 0$ must be satisfied with equality. Additionally, as $\xi_i = 0$, we have

$$y_i \left(\hat{\theta} \cdot \phi(x_i) + \hat{\theta}_0 \right) = 1.$$

Thus,

$$\hat{\theta}_0 = y_i - \hat{\theta} \cdot \phi(x_i) = y_i - \sum_{j=1}^n \hat{\alpha}_j y_j K(x_j, x_i).$$

In practice, we often average over all such support vectors:

$$\hat{\theta}_0 = \frac{1}{|S|} \sum_{i \in S} \left(y_i - \sum_{j=1}^n \hat{\alpha}_j y_j K(x_j, x_i) \right),$$

where $S = \{i \mid 0 < \hat{\alpha}_i < C\}$.

Definition of Support Vectors: A training example (x_i, y_i) is called a *support vector* if its corresponding Lagrange multiplier is non-zero, i.e.,

$$\hat{\alpha}_i > 0.$$

Moreover, the Karush-Kuhn-Tucker (KKT) complementary slackness condition states that

$$\hat{\alpha}_i \left[1 - y_i \left(\hat{\theta} \cdot \phi(x_i) + \hat{\theta}_0 \right) - \xi_i \right] = 0.$$

Thus, if $\hat{\alpha}_i > 0$, then necessarily

$$y_i \left(\hat{\theta} \cdot \phi(x_i) + \hat{\theta}_0 \right) = 1 - \xi_i.$$

Plugging in $\hat{\theta}$, we have:

$$y_i \left(\sum_{j=1}^n \hat{\alpha}_j y_j K(x_j, x_i) + \hat{\theta}_0 \right) = 1 - \xi_i.$$

- If $0 < \hat{\alpha}_i < C$, then $\mu_i > 0$, as derived from the stationary condition. According to the complementary slackness condition which states that $\mu_i \xi_i = 0$, $\xi_i = 0$ must be satisfied. Hence, for the point lies exactly on the margin, we have:

$$y_i \left(\sum_{j=1}^n \hat{\alpha}_j y_j K(x_j, x_i) + \hat{\theta}_0 \right) = 1.$$

- If $\hat{\alpha}_i = C$, then $\mu_i > 0$. In this case, $\xi_i > 0$, meaning the point is either misclassified or lies inside the margin, according to the primal feasibility constraints, we have:

$$y_i \left(\sum_{j=1}^n \hat{\alpha}_j y_j K(x_j, x_i) + \hat{\theta}_0 \right) \leq 1.$$